

Closed-Lie resolution of the higher-dimensional homogeneous-kernel problem

Conformal-symplectic existence in even dimension and odd-dimensional impossibility

candidate full closed-Lie solution—likely valid

1. Source problem and scope

Avetisyan and Karapetyants ask, at the end of Section 4.5 of arXiv:2112.03840v1 [1], for a natural subgroup of $GL^+(n)$, $n > 2$, producing an essentially unique homogeneous integral kernel analogous to their two-dimensional kernel, and for properties of the resulting operator. A short source excerpt is:

“what would be a natural choice of a subgroup $G \subset GL^+(n) \dots$ that produces an essentially unique homogeneous integral kernel”

The full statement is visible in the page-31 evidence crop below. The same open problem is explicitly reiterated in arXiv:2504.04127v1 [2], which studies the two-dimensional operator in function spaces.

As we have already mentioned in the Introduction, the operator (35) is of interest on its own side and we plan to study its properties in detail in another paper. It is obvious that the image of this operator is always a homogeneous function of order -1 , so it cannot be bounded, say, in Lebesgue L^p spaces. The study of this operator in another type of spaces can be a natural subject for investigation.

However, there is one more general question inspired by the content of this subsection, which we will formulate as an open problem.

Open problem: As noted above, in the case $n > 2$ there are no non-trivial kernels homogeneous with respect to all (orientation preserving) linear transformations. However, undoubtedly, the narrowing down of the group of transformations will give nontrivial kernels with certain characteristics. An interesting question therefore is: what would be a natural choice of a subgroup $G \subset GL^+(n)$ for $n > 2$ that produces an essentially unique homogeneous integral kernel K similar to the case $n = 2$ above, and what properties does the integral operator with that kernel possess?

5 Conclusion

The paper introduces, investigates and systematizes the concept of a homogeneous operator in a general context. Conditions for the homogeneity of the integral operator are given and questions of

We give a complete answer in the natural category of closed linear Lie subgroups acting transitively on the punctured space. In every even dimension the positive conformal-symplectic group gives the desired nontrivial Case-B mechanism. In odd dimension, Kramer’s classification of transitive linear groups [4] shows that no nonzero essentially unique Case-B kernel can exist: the only essentially unique kernel is the constant, Case-A kernel for $SL(n)$. Thus the apparent odd-dimensional gap is an obstruction, not a missing construction.

2. Definitions

For the constructive half, let $V = \mathbb{R}^{2m}$, $m \geq 2$, and fix a nondegenerate alternating form ω on V . Define the positive conformal-symplectic group

$$\mathrm{CSp}^+(V, \omega) := \{g \in GL(V) : \omega(gx, gy) = c(g)\omega(x, y) \text{ for some } c(g) > 0\}.$$

Writing $g = c(g)^{1/2}s$ shows that $s \in \mathrm{Sp}(V, \omega)$ and

$$\det g = c(g)^m > 0. \tag{1}$$

Thus $G := \mathrm{CSp}^+(V, \omega)$ is a natural subgroup of $GL^+(2m)$ and acts by dilations of Lebesgue measure with character $\lambda_g = \det g$.

Following [1], a measurable kernel K is strongly G -homogeneous when

$$K(gx, gy) = (\det g)^{-1}K(x, y) \tag{2}$$

for all $g \in G$, modulo the usual almost-everywhere identification. We call K symmetric or antisymmetric when $K(y, x) = K(x, y)$ or $K(y, x) = -K(x, y)$, respectively.

3. Proof intuition

The symplectic product is the missing relative invariant. The group G rescales it by $c(g)$ while Lebesgue measure rescales by $c(g)^m$. Consequently a kernel of weight $(\det g)^{-1}$ must have degree $-m$ in $\omega(x, y)$. Moreover, G is transitive on pairs with a fixed sign of $\omega(x, y)$; hence only two constants, one for each sign, survive. Swap symmetry relates those constants and leaves a one-dimensional class, just as in the source's orientation-preserving two-dimensional example.

The higher-order singularity is the analytic subtlety. The parity-matched function t^{-m} has a canonical homogeneous finite-part distribution, obtained by differentiating $\mathrm{pv}(1/t)$. Pulling it back through $t = \omega(x, y)$ gives the operator. This also reveals why the direct antisymmetric analogue behaves best for odd m : exactly then t^{-m} is antisymmetric.

Odd dimensions are controlled by classification rather than construction. Kramer's theorem leaves only similarity groups, $SL(d)$ and its scalar extensions, and the exceptional similarity group $G_2\mathbb{R}_{>0}$ in dimension seven. Similarity groups preserve continuous Gram parameters and therefore admit infinitely many kernels; $SL(d)$ is generically two-frame transitive and leaves only constants; adjoining any nontrivial scalar makes the determinant covariance incompatible with that constant and forces the kernel to vanish. These three elementary orbit calculations exhaust the odd-dimensional problem.

4. Generic-orbit classification and essential uniqueness

Lemma 1 (generic pair orbits). *The group $G = \mathrm{CSp}^+(V, \omega)$ acts transitively on $V \setminus \{0\}$. Its diagonal action on*

$$U = \{(x, y) \in V \times V : \omega(x, y) \neq 0\}$$

has exactly two orbits, $U_+ = \{\omega(x, y) > 0\}$ and $U_- = \{\omega(x, y) < 0\}$.

Proof. For $x \neq 0$, choose z with $\omega(x, z) = 1$ and extend (x, z) to a symplectic basis. Mapping one such basis to another proves that $\text{Sp}(V, \omega)$, hence G , is transitive on nonzero vectors.

Now take $(x, y), (x', y') \in U$ with $\omega(x, y)$ and $\omega(x', y')$ of the same sign, and put $c = \omega(x', y')/\omega(x, y) > 0$. The prescribed map $x \mapsto x', y \mapsto y'$ is c -conformal on the symplectic plane $\text{span}\{x, y\}$. Choose symplectic complements of this plane and its target and extend the map there by $c^{1/2}$ times a symplectic isomorphism. The resulting element of G sends (x, y) to (x', y') . The sign cannot change because $c(g) > 0$. $\square$

Theorem 2 (even-dimensional kernel classification). *Modulo a null set, every measurable kernel satisfying (2) has the form*

$$K_{A,B}(x, y) = \begin{cases} A |\omega(x, y)|^{-m}, & \omega(x, y) > 0, \\ B |\omega(x, y)|^{-m}, & \omega(x, y) < 0, \end{cases} \quad (3)$$

for unique constants A, B . Conversely, every such kernel is strongly homogeneous. Hence the space of homogeneous kernels is two-dimensional.

The symmetric and antisymmetric subspaces are each one-dimensional: symmetry is equivalent to $B = A$, while antisymmetry is equivalent to $B = -A$. In particular, the parity-matched kernel

$$K_m(x, y) = \omega(x, y)^{-m} \quad (4)$$

spans the unique class satisfying $K_m(y, x) = (-1)^m K_m(x, y)$. It is antisymmetric exactly when m is odd, i.e. when $\dim V \equiv 2 \pmod{4}$.

Proof. The hypersurface $\{\omega = 0\}$ has Lebesgue measure zero in $V \times V$. By Lemma 1, a homogeneous kernel is determined on U_+ and U_- by its values at pairs with symplectic product 1 and -1 . If p lies in either orbit and $t = \omega(p)$, an element taking the corresponding normalized pair to p has multiplier $|t|$. Equations (1) and (2) therefore force the factor $|t|^{-m}$, giving (3). Direct substitution proves the converse. Swapping the variables exchanges the two sign orbits, so the assertions about A, B and parity follow. $\square$

For $m = 1$, $G = GL^+(2)$ and Theorem 2 recovers exactly the two-dimensional orientation-split kernel space in [1]; its antisymmetric class is $1/\omega(x, y)$. Thus the construction is not merely analogous at the level of a formula: it extends the same orbit mechanism.

5. A canonical finite-part operator

Let

$$u_m := \text{Pf} \frac{1}{t^m} := \frac{(-1)^{m-1}}{(m-1)!} D^{m-1} \left(\text{pv} \frac{1}{t} \right). \quad (5)$$

Then u_m agrees with t^{-m} off the origin, is homogeneous of degree $-m$ under positive dilations, and has parity $u_m(-t) = (-1)^m u_m(t)$. For a Schwartz function ϕ ,

$$\langle u_m, \phi \rangle = \frac{1}{(m-1)!} \text{pv} \int_{\mathbb{R}} \frac{\phi^{(m-1)}(t)}{t} dt. \quad (6)$$

For $x \neq 0$, the linear functional $\ell_x(y) = \omega(x, y)$ is a submersion. Define

$$(T_m f)(x) := \langle u_m(\ell_x(\cdot)), f \rangle, \quad f \in \mathcal{S}(V). \quad (7)$$

Equivalently, let the one-direction Radon slice be

$$(R_x f)(t) = \int_{\ell_x(y)=t} \frac{f(y)}{\|\nabla \ell_x\|} d\sigma_t(y).$$

The coarea formula and (6) give the rigorous representation

$$(T_m f)(x) = \frac{1}{(m-1)!} \text{pv} \int_{\mathbb{R}} \frac{(R_x f)^{(m-1)}(t)}{t} dt. \quad (8)$$

This is a higher-order Radon/Hilbert formula. When $m = 1$ it reduces to the principal-value mechanism of the source's two-dimensional operator.

Proposition 3 (operator properties). *The operator (7) has the following properties.*

1. *It maps $\mathcal{S}(V)$ to smooth functions on $V \setminus \{0\}$ and is nonzero.*
2. *With $L_g f(x) = f(g^{-1}x)$, it is G -homogeneous:*

$$T_m L_g = L_g T_m \quad (g \in G).$$

3. *Every output is homogeneous of degree $-m$:*

$$(T_m f)(rx) = r^{-m} (T_m f)(x) \quad (r > 0).$$

Consequently, if $T_m f \neq 0$, then $T_m f \notin L^p(V)$ for every $1 \leq p \leq \infty$.

4. *On compactly supported smooth test functions away from the origin, its distributional kernel has formal-adjoint parity*

$$T_m^* = (-1)^m T_m.$$

Thus it is formally skew-adjoint for odd m and formally self-adjoint for even m .

Proof. The pullback of a distribution by a submersion is well defined, and its dependence on the nonzero parameter x is smooth. To see nonvanishing directly, fix $x_0 \neq 0$ and symplectic coordinates (a, t, w) with $\omega(x_0, y) = t$. Choose a Schwartz function $q(a, w)$ of integral 1 and put $f(a, t, w) = q(a, w) t^m e^{-t^2}$. The singularity cancels ordinarily and $(T_m f)(x_0) = \int_{\mathbb{R}} e^{-t^2} dt \neq 0$.

For covariance, substitute $y = gz$ in (7). Since

$$\omega(x, gz) = c(g) \omega(g^{-1}x, z), \quad u_m(c(g)t) = c(g)^{-m} u_m(t),$$

the factor $c(g)^{-m}$ cancels the Jacobian $\det g = c(g)^m$, leaving $(T_m f)(g^{-1}x)$. Separate scaling in the first argument of ω proves output homogeneity.

If a nonzero smooth function h on $V \setminus \{0\}$ is homogeneous of degree $-m$, its polar L^p integral contains

$$\int_0^\infty r^{2m-1-mp} dr,$$

which diverges at infinity for $p < 2$, at the origin for $p > 2$, and logarithmically at both ends for $p = 2$. For $p = \infty$, nonzero homogeneity forces blow-up at the origin. Finally, $\omega(y, x) = -\omega(x, y)$ and the parity of u_m give the formal-adjoint identity after swapping variables. $\square$

6. Ordinary principal values and the parity obstruction

The finite part in (7) is genuinely needed for $m > 1$. For fixed $x \neq 0$, the ordinary symmetric truncation

$$\lim_{\varepsilon \downarrow 0} \int_{|\omega(x,y)| > \varepsilon} \frac{f(y)}{\omega(x,y)^m} dy \quad (9)$$

exists if and only if

$$(R_x f)^{(j)}(0) = 0 \quad \text{for every } 0 \leq j \leq m-2 \text{ with } j \equiv m \pmod{2}. \quad (10)$$

Indeed, Taylor expansion of $R_x f(t)$ at zero shows that precisely these terms produce even, nonintegrable powers in the symmetric truncation; all opposite-parity singular terms cancel. Under (10), (9) equals the finite-part value.

There is also a sharp one-dimensional obstruction explaining the modulo-4 distinction. Let

$$k_{A,B}(t) = \begin{cases} A|t|^{-m}, & t > 0, \\ B|t|^{-m}, & t < 0. \end{cases}$$

Among scalar pullback regularizations $u(\omega(x,y))$, an exactly homogeneous distributional extension of degree $-m$ exists only if

$$B = (-1)^m A. \quad (11)$$

To prove this, for $\Re z > -1$ consider

$$\langle F_z, \phi \rangle = \int_0^\infty r^z (A\phi(r) + B\phi(-r)) dr.$$

Its meromorphic continuation has at $z = -m$ a residue proportional to

$$(A + (-1)^{m-1} B) \delta^{(m-1)}.$$

If this coefficient is nonzero, the finite part has a logarithmic scaling anomaly. The anomaly cannot be removed by adding a distribution supported at zero: for the Euler operator $E = tD$,

$$(E + m) \delta^{(j)} = (m - j - 1) \delta^{(j)},$$

so the obstructing $\delta^{(m-1)}$ coefficient is unchanged. This proves necessity; (5) proves sufficiency. Requiring the parity $(-1)^m$ also removes the otherwise possible homogeneous contact term $c\delta^{(m-1)}$, so u_m is canonical.

Consequently, for odd m the unique antisymmetric measurable class is precisely the class with a canonical homogeneous scalar finite part. For even m , the canonical scalar finite-part class is symmetric, while the antisymmetric measurable class requires either the cancellation domain appropriate to its stronger singularity or a regularization with a scaling anomaly. This is an analytic limitation, not a failure of the kernel classification.

7. Odd-dimensional classification and impossibility

We now write $d \geq 3$ for an odd integer and $M = \mathbb{R}^d \setminus \{0\}$. The next theorem treats every closed subgroup $G \subset GL^+(d)$ acting transitively on M , not merely connected ones.

Theorem 4 (odd-dimensional closed-Lie obstruction). *Let $d \geq 3$ be odd, and let $G \subset GL^+(d, \mathbb{R})$ be closed and transitive on M . Up to linear conjugacy, exactly one of the following kernel alternatives occurs.*

1. $G = SO(d)\mathbb{R}_{>0}$, or $d = 7$ and $G = G_2\mathbb{R}_{>0}$. These are Case-A groups and their homogeneous measurable kernel spaces are infinite-dimensional.
2. $G = SL(d, \mathbb{R})$. This is Case A and the homogeneous kernels are exactly the constants modulo null sets; the associated operator is

$$Tf(x) = C \int_{\mathbb{R}^d} f(y) dy.$$

3. $SL(d, \mathbb{R}) \subsetneq G \subseteq GL^+(d, \mathbb{R})$, including $G = GL^+(d, \mathbb{R})$. Then the only homogeneous measurable kernel is zero. These are Case-B groups.

Consequently, no closed transitive subgroup of $GL^+(d)$ in odd dimension produces a nonzero essentially unique Case-B kernel. The constant $SL(d)$ kernel is the sole essentially unique odd-dimensional alternative, but it is the degenerate Case-A solution already reduced by the source theory to invariant measure.

Proof. Because M is connected, the identity component G° is transitive. Indeed, G° has the same Lie algebra as G , so each G° -orbit has full dimension and is open; normality of G° makes all such orbits open, and connectedness permits only one.

Kramer's Theorem 6.17 [4] classifies closed connected linear groups transitive on nonzero real vectors. In odd dimension its list reduces to

$$SO(d)\mathbb{R}_{>0}, \quad SL(d, \mathbb{R}), \quad GL^+(d, \mathbb{R}),$$

together with $G_2\mathbb{R}_{>0}$ when $d = 7$. The normalizer table in Kramer's Theorem 6.14 shows that the first and exceptional groups have no larger subgroup inside $GL^+(d)$ with the same identity component. If $G^\circ = SL(d)$, every $g \in G$ can be written $g = rs$ with $r > 0$ and $s \in SL(d)$; since $s \in G$, the scalar rI also lies in G . Thus the remaining disconnected possibilities are precisely scalar extensions between $SL(d)$ and $GL^+(d)$.

It remains to compute kernels in these cases. For the similarity group, simultaneous rotation and positive scaling give all kernels

$$K(x, y) = |x|^{-d} F\left(\frac{|y|}{|x|}, \frac{\langle x, y \rangle}{|x||y|}\right), \quad (12)$$

where F is an arbitrary measurable function. Conversely every expression (12) is homogeneous. The formula is exact modulo the null coordinate sets because $SO(d)$ is transitive on pairs with a fixed Gram matrix when $d \geq 3$. Hence this space is infinite-dimensional. Since $G_2\mathbb{R}_{>0} \subset SO(7)\mathbb{R}_{>0}$, all the kernels in (12) remain homogeneous for the exceptional group, so its kernel space is also infinite-dimensional. In either similarity case, fixing a nonzero vector forces the scalar factor to be one, so the stabilizer has determinant one and the group is Case A.

For $SL(d)$, the set of linearly independent pairs has full measure, and $SL(d)$ is transitive on it: extend each pair to a basis and adjust one complementary basis vector to make the change-of-basis determinant one. Since the determinant character is trivial, every homogeneous kernel is therefore constant almost everywhere.

Finally suppose G properly contains $SL(d)$. Then $rI \in G$ for some $r \neq 1$. $SL(d)$ -invariance makes a kernel constant, say C , on the generic pair orbit, whereas covariance under rI gives

$$C = K(rx, ry) = r^{-d}K(x, y) = r^{-d}C.$$

Thus $C = 0$. Moreover, after fixing $x \neq 0$, choose $s \in SL(d)$ with $sx = r^{-1}x$; then rs fixes x and has determinant $r^d \neq 1$, so this is Case B. The three alternatives are exhaustive. $\square$

Corollary 5 (closed-Lie resolution of the source problem). *Within closed transitive linear Lie groups, the higher-dimensional problem has a complete answer. For even $d = 2m$, $G = \mathrm{CSp}^+(2m, \mathbb{R})$ gives the nonzero two-orientation kernel space in Theorem 2, with a unique symmetric and a unique antisymmetric class and the finite-part operator properties proved above. For odd d , Theorem 4 rules out every nonzero essentially unique Case-B kernel; only the constant $SL(d)$ Case-A operator remains.*

8. Verification and novelty boundary

The even-dimensional proof is self-contained apart from standard distribution pullback and coarea facts. The odd-dimensional exhaustion relies decisively on Kramer’s published Theorems 6.14 and 6.17 [4]; the deduction from the list is proved above. The orbit argument was checked explicitly in the low-dimensional cases $m = 1, 2, 3$, and the accompanying script verifies determinant, covariance, and swap parity for random elementary conformal-symplectic matrices. Computation is only a sanity check, not part of the proof.

A bounded search used the run’s four lightweight indexes; local arXiv source metadata; arXiv-id, title, author, “conformal symplectic”, “homogeneous integral kernel”, “transitive on nonzero vectors”, Kramer, and $GL^+(n)$ keyword combinations; and close-variant web searches. No paper was found that explicitly ties the conformal-symplectic construction or Kramer’s classification to the open problem in [1]. However, Clerc–Kobayashi–Ørsted–Pevzner [3] study kernels given by powers of the standard symplectic form, their homogeneous-distribution continuation, and associated intertwining operators on spheres. Therefore this packet does *not* claim novelty for symplectic-form power kernels, their general distribution theory, or the transitive-group classification itself. The candidate contribution is their exact synthesis in the source framework: the CSp^+ generic-orbit theorem and operator, plus the exhaustive odd-dimensional obstruction. A broader representation-theoretic review is still required before a publication-level novelty claim.

9. Scope boundary and further research

The extracted problem is resolved in the closed linear Lie category used by the source’s geometric setup. The following broader questions remain outside that claim:

- pathological nonclosed subgroups for which measurable invariance need not automatically extend to the closure;
- a function-space boundedness theory adapted to degree- m outputs;
- for dimensions divisible by 4, a comparably canonical antisymmetric operator without a scalar scaling anomaly;
- a comprehensive novelty comparison with symplectic Knapp–Stein and prehomogeneous-vector-space literature.

Human review should focus on the use of Kramer’s list and normalizers in Theorem 4, the orbit-extension step in Lemma 1, the distributional covariance calculation, and the residue obstruction. The current recommendation is to retain this as a likely-valid candidate full closed-Lie solution.

References

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